

Rohit Yadav

Built 25 LLMs from scratch • 120M–1.4B params • TPU clusters • Trained on 500 Billion+ tokens

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EDUCATION

National Institute of Technology, Jalandhar

Bachelor of Technology in Information Technology

Jalandhar, India

Aug 2023 – Aug 2027

Green Valley School for Children

CBSE Board 12th

Gandhinagar, India

May 2021 – May 2023

EXPERIENCE

Full Stack Intern — StudentVerse (Dubai)

Aug 2026 – Present

Product Engineering Intern — Remote

- Building the responsive web and mobile application screens and integrating APIs for student discovery.
- Deploying application services to the cloud using Docker containers to maintain stable uptime and fast load times.
- Managing cloud deployment and infrastructure hosting on Microsoft Azure for scalable, low-latency app delivery.

AI Research Intern — Molsys Pvt. Ltd.

Jun 2026 – Jul 2026

AI/ML Research Internship (Paid Contract Work) — Remote

- Built and fine-tuned custom LLM pipelines focusing on supervised fine-tuning, model distillation, and fast inference.
- Deployed large-scale models like Qwen on 2x NVIDIA H200 GPUs using llama.cpp server, Unsloth Studio, and vLLM for optimized, high-throughput inference.
- Engineered agentic AI workflows and set up production-ready model deployments using Amazon Bedrock.

Research Lead — AutonomousX

Apr 2026 – Present

Open Source LLM Research Organization

- Secured Google TPU Research Cloud (TRC) grant access to run large-scale model pretraining experiments.
- Trained 25 LLMs ranging from 120M to 1.4B parameters across 500B+ tokens using v4 and v6e TPU pods.

Amazon ML Summer School 2026 — Selected Cohort Participant

Jun 2026 – Jul 2026

Amazon — Machine Learning Training Program

- Selected among 3,000 students from over 134,000 applicants across India for intensive ML systems training.
- Completed hands-on coursework on foundational ML algorithms, Generative AI, LLMs, and distributed model optimization.

Open Source Contributor

Mar 2026 – Present

Meta LLaMA & Google DeepMind Gemma

- Submitted a pull request to **Google DeepMind Gemma** fixing tokenizer normalization and strict JAX typing issues.
<https://github.com/google-deepmind/gemma/pull/624>
- Submitted an upstream fix for **Meta LLaMA** addressing a broken CLI import that caused runtime download failures.
<https://github.com/meta-llama/llama-models/pull/476>
- Actively engaged in open-source production-grade foundation model repositories.

Full-Stack Web Developer Intern

Jun 2025 – Jul 2025

Skylark Express Delhi Pvt Ltd

On-site, Gurugram

- Built an internal operations dashboard from scratch to track package movement and simplify day-to-day logistics.
- Redesigned frontend views to make data tables load faster and render properly across desktop and mobile screens.

PROJECTS

AutonomousX: Built 25 LLMs From Scratch (0.1B–1.4B) on Google TPUs

<https://huggingface.co/autonomousX>

JAX, Flax, Orbx, Google Cloud TPUs (v4/v6e), Distributed Systems, Scaling Laws

- Engineered an open-source research initiative building 25 language models (124M to 1.4B parameters) completely from scratch on Google Cloud TPU pods under the awarded **Google TRC grant**.
- Implemented distributed training pipelines in JAX/Flax with Orbx checkpointing across TPU v4-8 pods (128GB HBM each) and v6e pods across 6 custom architectures, testing scaling laws, optimizers (Lion vs. AdamW), positional encodings (RoPE vs. learned), and datasets (Dolma, PILE) over 500B+ tokens.
- Open-sourced all raw model weights, Orbx checkpoints, training loss/perplexity curves, and a 10-minute reproducible Colab notebook for instant inference and community verification.

GiniVibe: Full-stack End-to-End Scalable Social Discovery Platform

<https://github.com/YADAV1825/Ginivibe-project>

Next.js 16, React Native (Expo), Node.js, Express, Redis, Kafka, Azure, Docker, PostgreSQL, WebRTC

- Developed an end-to-end social network across web and mobile (iOS & Android) featuring multimedia feeds, encrypted chat, online/offline event hubs, group audio/video rooms, AI personas, and personality insights.
- Designed a sleek Neumorphic UI and a hybrid backend combining a monolithic core API with microservices for matchmaking and events, backed by PostgreSQL, Prisma, and Docker containerization.
- Architected for million-user scale with Redis, Kafka, and Azure cloud deployment, featuring an intent-driven AI matcher where natural prompts (e.g., “connect with a founder in Delhi”) discover and rank users instantly.

PathoPreter: DNA Foundation Model for Clinical Genetics

https://huggingface.co/autonomousX/PathoPreter-DNA-Pathogen-Clinvar_gnomAD_ranker

Python, PyTorch, Nucleotide Transformer (MLM), Custom MLP Head, SHAP, Clinical Genomics

- Engineered a clinical AI model that analyzes human DNA mutations to predict whether rare genetic variants cause disease.
- Constructed on a 500M-parameter Nucleotide Transformer masked language model (MLM) backbone with a custom multi-layer perceptron (MLP) classification head, fine-tuned on a leakage-free dataset of 1.25M curated genomic variants.
- Built 80x smaller than EVO2 while identifying 75%+ of harmful mutations by testing just 10% of variants. Achieved 0.9186 ROC-AUC on a 14K rare-variant benchmark while identifying 75%+ of harmful mutations by testing only 10% of variants.

Large-Scale Genomic Data Pipeline (1.25M Variants)

https://github.com/YADAV1825/PathoPreter/tree/main/data_processing

Python, Multiprocessing, Pandas, NumPy, ClinVar, gnomAD

- Built a high-speed data engineering pipeline that cleans and formats massive raw DNA records for machine learning.
- Developed 32-core parallel processing scripts to merge 7.7M raw records from multiple global gene databases into unified binary files.
- Reduced dataset processing time from over 150 hours down to under 20 hours while producing zero-leakage, GPU-ready shards.

RAGE-BORN: Agentic CLI Harness & Browser-Aware Agent

<https://github.com/YADAV1825/RAGE-BORN>

Python, LLM APIs (Claude, Gemini), Playwright, Linux CLI, Bash Automation

- Engineered an autonomous developer agent harness (similar to Claude Code and OpenCode) that operates directly inside the terminal and browser to execute tasks through natural language.
- Designed a planner-execution feedback engine orchestrating 25+ local CLI tools, file system operations, and multi-model LLM APIs.
- Implemented browser-aware automation that visibly launches a live browser, navigates sites, fills forms, conducts research, and syncs findings directly to Google Workspace in real time.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++, SQL, HTML, CSS

Frontend & Mobile: Next.js 16, React 19, React Native, Expo 57, Tailwind CSS, Reanimated, Zustand

Backend & Cloud: Node.js, Express.js, WebSockets, Socket.io, WebRTC, Docker, AWS, GCP, Azure

AI & Distributed Training: LLM Pretraining & Fine-tuning, TPU (v4/v6e) & GPU (A100/H100) Clusters, JAX pmap, PyTorch, HuggingFace

Databases & DevOps: PostgreSQL, Prisma ORM, MongoDB, Git, Linux, CI/CD

ACHIEVEMENTS

Awarded Google TRC grant for 320 TPU access (SPOT + STANDARD VMs).

Awarded AMD MI300X GPU access for 300 non-preemptive hours

IOQM (PRMO): Qualified for Indian Olympiad Qualifier in Mathematics

KVPY SA: AIR 2590

LeetCode: Knight (2013), 600+ problems solved across all platforms

CodeChef: 4-Star (1818)

Codeforces: Specialist (1417)

CodeChef START166B: Global Rank 9

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